

9.1 Relations and Their Properties

Monday, November 18, 2024 1:21 PM

A relation is generally something we define to compare elements in a set.

Common relations we use are ...

$=, \leq, <, \subseteq, |, \equiv \pmod{n}$

Definition: Let A and B be sets. A

binary relation from A to B is a subset of $A \times B$

So $A \times B = \{(a, b) \mid a \in A \wedge b \in B\}$ where every tuple (a, b) tells us that "a is related to b."

We let R be the binary relation set.

If $(a, b) \in R$, we may also write aRb (or $a \not R b$ if $(a, b) \notin R$).

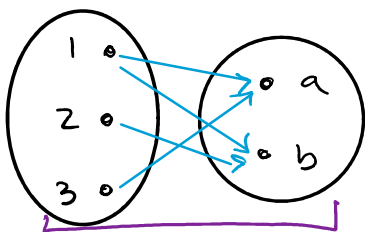
Let $A = \{\text{cities in the U.S.}\}$, $B = \{\text{states in the U.S.}\}$.

If R is the relation aRb only if "a" is a city in "b," what are some elements of R ?
(Milwaukee, WI), (Fiona City, IA), (St. Louis, MO)
(San Francisco, CA), ...

There are several ways to visualize a relation.

If $A = \{1, 2, 3\}$, $B = \{a, b\}$, and

$R = \{(1, a), (1, b), (3, a), (2, b)\}$ then...



or

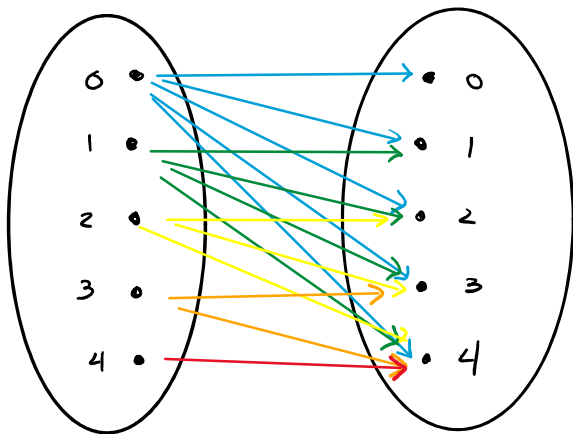
R	a	b
1	✓	✓
2		✓
3	✓	

This looks a lot like how we visualize functions. Can R be used to define a function? What about the reverse, can a function define a relation?

Since $a \mapsto 1$ and $a \mapsto 3$, R can't be used to define a function. If we had some function $f: A \rightarrow B$, the range of f is a subset of $A \times B$ with elements $(a, f(a))$, so any function does define a relation.

Definition: A relation on a set A is a relation from A to A (a subset of $A \times A$).

If $A = \{0, 1, 2, 3, 4\}$, what are all elements in $R = \{(a, b) \mid (a, b \in A) \wedge (a \leq b)\}$?



R	0	1	2	3	4
0	✓	✓	✓	✓	✓
1		✓	✓	✓	✓
2			✓	✓	✓
3				✓	✓
4					✓

If $|A| = n$, how many relations can we define from A to A ?

Since $R \subseteq A \times A$ and $|A \times A| = n^2$, there are 2^{n^2} total subsets of $A \times A$, thus 2^{n^2} total relations we can define from A to A .

If $|A|=n$, $|B|=m$ we know a relation from A to B is a subset of $A \times B$ where $|A \times B| = n \cdot m$, so there are $2^{n \cdot m}$ possible relations from A to B .

Some Properties of Relations

Definition: A relation on a set A is reflexive if $(a,a) \in R$ (aRa) for every $a \in A$.

Definition: A relation on a set A is symmetric if $(b,a) \in R$ whenever $(a,b) \in R$ ($aRb \Leftrightarrow bRa$) for all $a, b \in A$.

Definition: A relation on a set A is antisymmetric if when $(a,b) \in R$ and $(b,a) \in R$ then $a=b$ ($aRb \wedge bRa \Rightarrow a=b$) for all $a, b \in A$.

Although one term includes the prefix "anti," these are not opposites. R may be both, only one, or neither.

Definition: A relation on a set A is transitive if when $(a,b) \in R$ and $(b,c) \in R$ then $(a,c) \in R$ ($aRb \wedge bRc \Rightarrow aRc$) for all $a, b, c \in R$.

What properties do the following relations have on $\mathbb{Z}$? $=$, $\leq$, $<$

Equal: $a \in \mathbb{Z}$, then $a=a$ (T)

$a, b \in \mathbb{Z}$, then $a=b \Leftrightarrow b=a$ (T)

$a, b \in \mathbb{Z}$ then $a=b$ and $b=a \Rightarrow a=b$ (T)

$a, b, c \in \mathbb{Z}$, then $a=b \wedge b=c \Rightarrow a=c$ (T)

Less Than or Equal: $a \in \mathbb{Z}$, then $a \leq a$ (T)

Less Than or Equal: $a \in \mathbb{Z}$, then $a \leq a$ (T)

$a, b \in \mathbb{Z}$, then $a \leq b \Leftrightarrow b \leq a$ (F)

$a, b \in \mathbb{Z}$, then $a \leq b \wedge b \leq a \Rightarrow a = b$ (T)

$a, b, c \in \mathbb{Z}$, then $a \leq b \wedge b \leq c \Rightarrow a \leq c$ (T)

Less Than: $a \in \mathbb{Z}$, then $a < a$ (F)

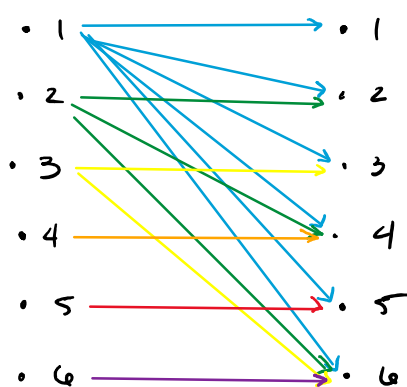
$a, b \in \mathbb{Z}$, then $a < b \Leftrightarrow b < a$ (F)

$a, b \in \mathbb{Z}$, then $a < b \wedge b < a \Rightarrow a = b$ (F)

$a, b, c \in \mathbb{Z}$, then $a < b \wedge b < c \Rightarrow a < c$ (T)

Find all elements of $R = \{(a, b) \mid (a, b \in A) \wedge (a|b)\}$

if $A = \{1, 2, 3, 4, 5, 6\}$.



Is this relation

reflexive? Yes (horiz. lines)

symmetric? No $2|4$ but $4 \nmid 2$

anti-symmetric? Yes $\overset{a}{2}|\overset{b}{2} \wedge \overset{b}{2}|\overset{a}{2} \Rightarrow a=b$

transitive? Yes $1|3 \wedge 3|6 \Rightarrow 1|6$ etc.

If $R = \{(a, b) \mid (a, b \in \mathbb{R}) \wedge (a+b=0)\}$, then is R

reflexive? symmetric? anti-symmetric? transitive?

Reflexive if $(a, a) \in R$ for all $a \in \mathbb{R}$.

$a+a=0 \Rightarrow a=-a$ only if $a=0$.

Symmetric if $(a, b) \in R$ then $(b, a) \in R$

$a+b=0 \Leftrightarrow b+a=0$ is true, addition is commutative.

Anti-symmetric if (a, b) and $(b, a) \in R \Rightarrow a=b$.

$a+b=0$ and $b+a=0$ are equivalent,

only makes $a=-b$ or $b=-a$.

Transitive if (a, b) and $(b, c) \in R \Rightarrow (a, c) \in R$

$$a+b=0 \text{ and } b+c=0 \Rightarrow b=-c \text{ so}$$

$$a-c=0 \text{ but this is not } a+c=0.$$

Only symmetric.

What if $R = \{(a,b) \mid (a,b \in \mathbb{R}) \wedge (a \cdot b \geq 0)\}$?

Reflexive if $(a,a) \in R$ for all $a \in A$.

$$a \cdot a \geq 0 \Leftrightarrow a^2 \geq 0 \text{ is true .}$$

Symmetric if $(a,b) \in R \Rightarrow (b,a) \in R$.

$a \cdot b \geq 0 \Leftrightarrow b \cdot a \geq 0$. Multiplication is commutative.

Anti-symmetric if $(a,b) \wedge (b,a) \in R \Rightarrow a=b$

$$a \cdot b \geq 0 \wedge b \cdot a \geq 0 \text{ does not force } a=b. (-1) \cdot (-2)$$

Transitive if $(a,b) \wedge (b,c) \in R \Rightarrow (a,c) \in R$.

$$\underline{a \cdot b \geq 0 \text{ and } b \cdot c \geq 0, \quad a \cdot c \stackrel{?}{\geq} 0}$$

If $a=0$

True

If $c=0$

True

If $b=0, a=1$

$b=0, c=-5$

$(1) \cdot (-5) \neq 0$

False!!

So R fails to be transitive

Reflexive, Symmetric